Aman Kumar

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Full-Stack AI Engineer · LangGraph · FastAPI · React · IIT Roorkee '26

PROFESSIONAL SUMMARY

IIT Roorkee graduate with strong full-stack and AI engineering fundamentals: agentic systems, retrieval-augmented generation, and backend API design. Comfortable working across the stack, from Python backends to React frontends, with a focus on reliable, production-grade LLM systems, structured output enforcement, self-checking agents, and gradable evaluation.

EDUCATION

Indian Institute of Technology, Roorkee — Roorkee, Uttarakhand
B.Tech + M.Tech (Integrated Dual Degree), Geophysical Technology — 70% — *2026*

Gaya College (BSEB) — Gaya, Bihar
Intermediate (Class XII) — 77.40% — *2021*

B.L Indo Anglian Public School (CBSE) — Gaya, Bihar
Matriculate (Class X) — 86.80% — *2019*

TECHNICAL SKILLS

AI & Agentic	LangGraph, LangChain, RAG, FAISS, BM25, Cross-encoder Reranking, HuggingFace, Groq (Llama 3.1), Anthropic Claude, Prompt Engineering, Structured Output, Vector Embeddings
Backend & APIs	Python, FastAPI, Flask, Node.js, NestJS (Express), REST API Design, JWT Auth, WebSockets, SSE, Pydantic, SQLAlchemy, Alembic, TypeORM
Frontend	React.js, Next.js (App Router), TypeScript, Redux Toolkit, JavaScript, Tailwind CSS
Databases	PostgreSQL, MySQL, MongoDB, SQLite, FAISS
DevOps & Tools	Git, GitHub, Docker, Vercel, Render, Postman, CI/CD concepts

WORK EXPERIENCE

Software Development Intern — Jun 2025 – Jul 2025
CallSure AI

- Built and deployed a LangChain + FAISS retrieval-augmented generation pipeline on Groq Llama, integrated into a live production platform to automate customer support resolution.
- Built and validated a HuggingFace-based classification model against a held-out test set, delivering all sprint work with clean, review-ready code and architecture documentation.

PROJECTS

Legal Document Intelligence: RAG Platform | FastAPI, React, FAISS, BM25 — Jan 2025 – Oct 2026

- Built a contract question-answering platform with clause-aware chunking and hybrid retrieval (FAISS + BM25 fused with RRF, cross-encoder reranking); answers stream over SSE with server-validated clause citations in 1 to 2 seconds.
- Wrote a benchmark comparing five retrieval strategies on 24 questions: hybrid reached 100% hit@5, reranking lifted hit@1 to 0.75 (MRR 0.85), and top-5 context used about 77% fewer prompt tokens than the full corpus.

Autonomous Code Review Agent | FastAPI, LangGraph, React, FAISS — Feb 2025 – Oct 2026

- Built a LangGraph multi-agent reviewer: static analysis and FAISS-retrieved guidance feed four parallel specialist agents, then a verifier drops findings on nonexistent lines and merges duplicates before scoring.
- Added GitHub PR review (changed-lines scope, HMAC-verified webhook, inline comments) and a live SSE pipeline view with a Monaco code viewer; reviews finish in 5 to 40 seconds.

AI-First CRM for Field Reps | FastAPI, LangGraph, React, PostgreSQL — Jun 2025 – Oct 2026

- Rebuilt a LangGraph CRM assistant so every AI write is a proposal the rep must confirm, and moved identity from the prompt to the JWT, closing an impersonation flaw in the first version.
- Added AI quick-log with an editable preview, rep and manager roles, at-risk doctor detection and analytics; covered by 10 automated tests.

LostFound: Campus Platform | FastAPI, React, WebSockets, Docker — Oct 2024 – Oct 2026

- Rebuilt a campus lost-and-found app, first deployed at IIT Roorkee, as a FastAPI and React platform with a match engine scoring text, category, place and timing, and real-time WebSocket notifications.
- Added secret-detail claim verification, JWT with refresh rotation, rate limiting, an admin audit log and Alembic migrations; 11 automated tests.

ACHIEVEMENTS

- Codeforces 1550 (Specialist) · CodeChef 2100 (5-Star) · 1500+ algorithmic problems solved.
- Top 1% in JEE Mains and JEE Advanced among 1.2M+ candidates, admitted to IIT Roorkee.